The University of Oxford

MSc (Computer Science / Mathematics and Foundations of Computer Science)

Applied Category Theory

Trinity Term 2026

*You have to choose exactly **one** of the following short projects.*

Option A: Codesign for a Satellite Constellation

A space agency is designing a satellite constellation. The design involves **three coupled subsystems**. We model this using **feasibility relations** (profunctors enriched over $\mathbf{Bool} = (\{0,1\}, \wedge, 1)$), given by the following tables.

- **Orbital Slots (D_{slot}):**

Slot	Altitude (km)	Radiation exposure	Debris risk	Launch windows
L1	550	Medium	0.003	Q1, Q2 (2)
L2	550	High	0.001	Q1 (1)
L3	400	Low	0.008	Q1, Q2, Q3 (3)
L4	650	Medium	0.002	Q2, Q3 (2)

- **Satellite Buses (D_{bus}):**

Config.	Mass (kg)	Power (W)	Downlink (Mbps)	Radiation tol.	Min. alt. (km)
μSat	25	40	10	Low	350
SmallSat-A	80	120	50	Medium	400
SmallSat-B	150	200	150	High	500

- **Payloads (D_{pay}):**

Instrument	Power (W)	Data (Mbps)	Mass (kg)	Science value
Multispectral	35	20	8	3
Hyperspectral	80	80	25	7
SAR	150	120	45	10
Thermal	25	15	12	4

Task 1: Identifying Codesign Structure

1. Identify the preorders of resources which appear in the problem.
2. Model the situation in terms of **Bool**-enriched profunctors. Which resources are required and which are provided?
3. Which composition (sequential, parallel) of the profunctors models the problem? Express it in terms of string diagrams.

Task 2: Feasibility Analysis

1. A **single-satellite configuration** is a triple $(s, b, p) \in D_{\text{slot}} \times D_{\text{bus}} \times D_{\text{pay}}$.(We say (s, b, p) is **feasible** if is mapped to 1 by the composite profunctor computed in the previous task.)

   List all feasible single-satellite configurations. Show your work for at least three configurations (one feasible, two infeasible with different failure modes).

2. Call a **convenient 3-satellite constellation** a multiset $\{(s_1, b_1, p_1), (s_2, b_2, p_2), (s_3, b_3, p_3)\}$ of feasible configurations satisfying:

   - Total science value ≥ 18
   - Total debris risk ≤ 0.012
   - All slots share at least one common launch window
   - Each slot is used at most once.

   Find all convenient 3-satellite constellations.

3. Among feasible constellations, identify the Pareto-optimal ones with respect to the pair (*maximize science value*, *minimize debris*). (A constellation C is **Pareto-optimal** if no other feasible C' has both weakly-better science and weakly-better debris with at least one strictly better.)

Task 3: Upgrades

The agency can invest in one upgrade:

- **Upgrade X:** SmallSat-A's radiation tolerance becomes High
- **Upgrade Y:** μSat's power becomes 70W

1. Compute the feasible configurations before and after each upgrade. Which upgrade adds more feasible single-satellite configurations?

2. Can you define these upgrades (and others) as morphisms of a convenient category?

Creative extension for full credit: Time-varying feasibility

Scenario: Satellites degrade over time. Each year:

- Power generation drops by 10%
- Debris risk at each slot increases by 0.001

Therefore, A constellation that is feasible at $t = 0$ may become infeasible at $t = T$.

Extend the framework to model **multi-period feasibility** where the agency can choose **reconfiguration actions** at each time step (e.g., decommission a satellite, launch a replacement).

1. Define a categorical structure for "time-indexed codesign problems." This should capture:

   - A family of feasibility relations Φ_t for each time t
   - Constraints between consecutive time steps (feasibility at t constrains what's possible at $t + 1$)

2. Apply your framework to the concrete scenario. Starting from a Pareto-optimal constellation at $t = 0$, compute the maximum T for which it remains feasible with no reconfiguration.

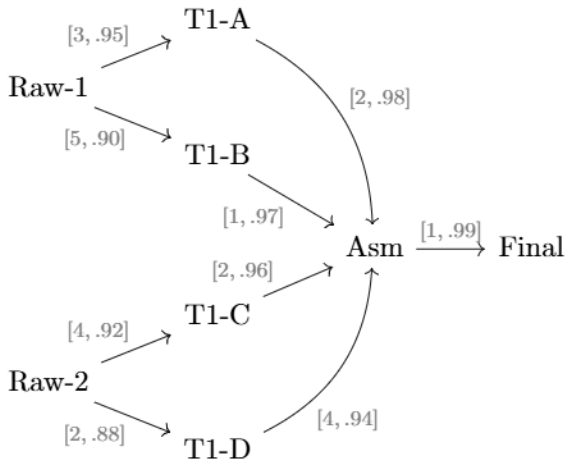
3. Discuss: What categorical structure would allow you to **optimize** over reconfiguration strategies? (You don't need to fully develop this—a precise problem statement and sketch of approach is sufficient.)

Option B: Enriched Categories for Supply Chain Risk

A manufacturer sources components through a multi-tier supply network. We model this using **an enrichment** capturing (time, reliability) pairs.

Let $\mathcal{Q}$ be the set of **Pareto-closed subsets** of $\mathbb{R}_{\geq 0} \times (0, 1]$ pairs, which we view as modeling (*time*, *reliability*), where reliability can be seen as a probability (of the component not failing). Here "Pareto-closed" means: if $(t, p) \in S$ and $t' \geq t$ and $p' \leq p$, then $(t', p') \in S$. Consider $\mathcal{Q}$ ordered by reverse inclusion: $S \leq S'$ iff $S \supseteq S'$ (more points = worse options = smaller in the order).

Consider now the following $\mathcal{Q}$-enriched graph, modeling a supply network.



(An edge label $[t, p]$ means Pareto-closure of $\{(t, p)\}$, i.e., $\{(t', p') : t' \geq t,\ p' \leq p\}$.)

Task 1: Monoidal structure

1. Equip $\mathcal{Q}$ with a monoidal structure which is convenient for the model at hand. Justify your claims.
2. Verify that the monoidal structure is compatible with the order (and if not, choose a different one).

Task 2: The Enriched Category of Supply Chains

1. Define the $\mathcal{Q}$-enriched category $\mathcal{C}$ as the one freely generated by the graph given above.
2. Compute these hom-objects (give the Pareto frontier of each):
   - $\mathcal{C}(\text{Raw-1}, \text{Asm})$
   - $\mathcal{C}(\text{Raw-2}, \text{Asm})$
   - $\mathcal{C}(\text{Raw-1}, \text{Final})$

   Show your work: trace all paths, compose, then find the frontier.

Task 3: Specifications

The manufacturer wants deliveries satisfying: **reliability** ≥ 0.90 **AND time** ≤ 8 **days**.

1. For each source node $X \in \{\text{Raw-1}, \text{Raw-2}\}$, determine whether the Pareto frontier of $\mathcal{C}(X, \text{Final})$ contains a point (t, p) satisfying $t \leq 8$ and $p \geq 0.90$. Which sources can meet the specification?

2. The specification defines an **up-set** (or **upper set**) $U = \{(t, p) : t \leq 8 \text{ and } p \geq 0.90\}$. Observe that U is *not* an element of $\mathcal{Q}$ (why?).

   One could try to define a "sub-hom-object" by

   $$\mathcal{C}(X, \text{Final})_{\text{spec}} \coloneqq \mathcal{C}(X, \text{Final}) \cap U.$$

   This captures "delivery options from X to Final that meet the spec."

   Does the assignment $X \mapsto \mathcal{C}(X, \text{Final})_{\text{spec}}$ define a $\mathcal{Q}$-enriched functor $\mathcal{C}^{\text{op}} \to \mathcal{Q}$? (That is: is $\mathcal{C}(-, \text{Final})_{\text{spec}}$ a $\mathcal{Q}$-enriched presheaf?) Justify your answer.

Creative Extension for full credit: Correlated Failures

Scenario: Suppliers T1-A and T1-C share a sub-supplier. If the sub-supplier fails, both T1-A and T1-C fail. Specifically:

- The sub-supplier fails with probability 0.15
- When it fails, both edges into T1-A and both edges into T1-C fail
- When it doesn't fail, all edges behave independently with their given reliabilities

Extend the categorical framework to handle correlated failures.

1. Explain why the our enrichment in $\mathcal{Q}$, alone, cannot express this correlation. What aspect of the monoidal structure breaks?
2. Propose an extended framework which could model correlations. Define your construction precisely. You need not develop the full theory, but definitions should be mathematically complete.
3. Apply your framework to compute the "effective" hom-object $\mathcal{C}(\text{Raw-1}, \text{Final})$ under the correlated failure model. Compare to the independent case from Task 2.

- Please typeset your project using LaTeX or a similar system. Diagrams and string diagrams in LaTeX can be created using interfaces such as *Quiver* (`https://q.uiver.app`) and *TikZit* (`https://tikzit.github.io/`), but you can also draw them manually.
- Make sure to cite any sources you use, including blogs or websites such as the nLab.
- If you write code to help you with the computations or the combinatorics, please provide it.
- Feel free to use any AI tools to help you, but use your critical thinking: hallucinations in category theory are still quite frequent. Check independently any mathematical assertions. Keep also in mind that the output of an AI tool is not an acceptable reference to cite.